

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an energy equation between PE and EPE	AO3.3	M1	$0.5 \times 2 \times 10 = \frac{\lambda}{2 \times 0.2} \times 0.4^2$ $\lambda = \frac{10}{0.4}$ $= 25 \text{ Nm}^{-1}$ $2 \times 10 = \frac{25}{0.2} e$ $e = \frac{2 \times 10 \times 0.2}{25}$ $= 0.16$ $2 \times 10 \times 0.26 = \frac{1}{2} \times 2v^2$ $+ \frac{25 \times 0.16^2}{2 \times 0.4}$ $v^2 = 3.6$ $v = 1.897 \dots$ $= 2.0 \text{ m s}^{-1} \text{ (to 1sf)}$
	Obtains correct value for λ or k	AO1.1b	A1	
	Using Hooke's Law to find extension at equilibrium	AO3.4	M1	
	Obtains correct extension at equilibrium using 'their' λ or k	AO1.1b	A1F	
	Forms equation using conservation of energy.	AO3.4	M1	
	Obtains the correct speed. (Only accept $V = 2 \text{ m s}^{-1}$ or $V = 2.0 \text{ m s}^{-1}$)	AO1.1b	A1	
(b)	States appropriate refinement.	AO3.5c	E1	Could include air resistance.
				Total 7 marks

Q2.

(a) using $EPE = \frac{\lambda x^2}{2l}$,

M1

$$EPE = \frac{32 \times 2.2^2}{2 \times 0.8}$$

B1 for 2.2

B1

$= 96.8 \text{ J}$

A1

3

- (b) by C of Energy, when next at rest, EPE (initial) = work done against friction + EPE (when at rest)

M1A1 for work done by friction or $5F$

M1A1

$$96.8 = F \times 5 + \frac{32 \times 1.2^2}{2 \times 0.8}$$

M1 3 terms; A1 all correct

M1A1

$$5F = 96.8 - 28.8$$

B1 28.8

B1

frictional force is 13.6N

A1

6

- (c) at B, tension is $\frac{32 \times 1.2}{0.8}$
 $= 48\text{N}$

B1

tension > friction
 hence particle starts to move

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E1

2

- (d) when particle is next at rest,
 work done against friction is EPE at B

$$13.6 \times \text{distance} = 28.8$$

M1

distance is 2.1176

$$= 2.12 \text{ m}$$

CAO

A1

2

- (e) total distance is $5 + 2.1176$
ft from M1 in (d)

or total distance $\times 13.6 = \text{original EPE, } 96.8$

B1

1

$$= 7.12 \text{ m}$$

total distance is 7.12 m

[14]

Q3.

(a) Initial EPE = $\frac{\lambda x^2}{2l}$

$$= \frac{120 \times (0.5)^2}{2 \times 5}$$

M1 for formula with extension 0.5

$$= 3 \text{ J}$$

M1

A1

Initial KE is $\frac{1}{2} \times 0.4 \times 9^2 = 16.2 \text{ J}$

When block is at A, $\frac{1}{2} mv^2 = 3 + 16.2$

M1

$$v^2 = 19.2 \div 0.2 = 96$$

Speed is 9.80 m s^{-1}

Accept $4\sqrt{6}$; condone 9.79

A1

4

(b) (i) Normal reaction is $mg = 0.4g$

M1

Frictional force is $0.4\mu g \text{ N}$

A1

Work done by frictional force is
 $5.5 \times (0.4\mu g)$ or $2.2\mu g$

m1

C of Energy, when at A, gives

$$19.2 - 5.5 \times (0.4\mu g) = \frac{1}{2} \times 0.4 \times v^2$$

Three terms, eg initial energy in (a)
(=3 or 19.2); work done; KE at A.

M1

Fully correct

A1

$$19.2 - 2.2\mu g = 0.2v^2$$

$$v = \sqrt{96 - 11\mu g}$$

$$Ft v = \sqrt{(v^2 \text{ in (a)}) - 11\mu g}$$

A1

6

(ii) Speed when rebounding is $\frac{1}{2} \sqrt{96 - 11\mu g}$

B1ft

Block is stationary at B

$$\frac{1}{2} \times 0.4 \times \frac{1}{4} (96 - 11\mu g) - 2.2\mu g$$

Three terms

M1

Two terms correct with sign

A1

$$= \frac{120 \times (0.5)^2}{2 \times 5}$$

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Third term correct with sign

A1

$$\frac{1}{2} \times 0.1(96 - 11\mu g) - 2.2\mu g = 3$$

$$4.8 - 2.75\mu g = 3$$

Or $4.8 - 0.55\mu g - 2.2\mu g = 3$

A1

$$\mu = 0.0668$$

A1

6

[16]

Q4.

- (a) At maximum speed,
tractive force = resistance force

M1

Using power = force × velocity:
 $800\,000 = F \times 40$

M1

$$F = 20\,000 \text{ N}$$

A1

3

- (b) Using force × distance = work done = change in energy:

$$20\,000 s = \frac{1}{2} \times 60\,000 \times (40^2 - 36^2)$$

M1 $Fs = \text{change of KE}$

M1

A1 2 of 3 terms correct

A1

A1 all 3 terms correct

A1

$$\text{Distance} = 456 \text{ m}$$

A1

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Q5.

(a)
$$\text{EPE} = \frac{\lambda x^2}{2l}$$
$$= \frac{180 \times 0.8^2}{2 \times 1.2}$$

M1

$$= 48 \text{ J}$$

A1

2

- (b) Using initial EPE = KE when string becomes slack:

M1

$$48 = \frac{1}{2} \times 5 \times v^2$$

A1F

$$v = \sqrt{\frac{96}{5}}$$

$$= 4.38 \text{ m s}^{-1}$$

$$ft \sqrt{\frac{a'}{2.5}}$$

A1F

3

(c) Normal reaction is 5g or 49

M1

Frictional force is $5g \times \mu$

m1 A1

Work done by frictional force is $5\mu g \times 2$

m1

$$= 10 \mu g$$

A1

Stops at wall $\Rightarrow 10\mu g = 48$
 $m1 \ 10 \mu g = 'a'$

m1

$$\mu = 0.490$$

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 accept $\frac{24}{49}$ OE

A1

7

[12]

Q6.

(a) EPE is $\frac{\lambda x^2}{2l}$

$$= \frac{300 \times 2^2}{2 \times 6}$$

M1

$$= 100 \text{ J}$$

A1

2

- (b) Using $F = \mu R$:
Friction is $0.3 \times 4g$

M1

$$= 1.2g \text{ or } 11.76 \text{ N}$$

A1

When string becomes slack:
Work done against friction = change in energy

m1

$$1.2g \times 2 = 100 - \frac{1}{2}mv^2$$

A1A1

$$v^2 = 38.245$$

Speed is 6.18 m s^{-1}

AG

A1

6

[8]

Q7.

Work done = change in PE

M1

$$= 800g \times 200$$

m1

$$= 1568000(\text{J})$$

Accept 3sf

A1

3

[3]

Q8.

(a) $\frac{1}{2} \times 35 \times v^2 = 35 \times 9.8 \times 10$

Energy method

M1A1

$$v = 14 \text{ (m s}^{-1}\text{)}$$

A1

3

(b) Air resistance or friction

B1

1

(c) Energy lost = $35 \times 9.8 \times 10 - \frac{1}{2} \times 35 \times 12^2$ (= 910)
Difference attempted

M1

$\pm$

A1

Work done: $F \times 20$ (= 910)

m1

$20F = 910$
 $F > 0$ $F = 45.5 \text{ (N)}$

A1

4

[8]

Q9.

(a) EPE is $\frac{\lambda x^2}{2l}$
 $= \frac{200(0.5)^2}{2 \times 2}$ Exam Papers Practice. All rights reserved.

M1

= 12.5 J

A1

2

(b) When string becomes slack,

using $\frac{1}{2}mv^2$ = loss in EPE

NB Using $\sqrt{5}$ to answer (a) and thus (b) $\Rightarrow$ no marks

M1

$\frac{1}{2} \times 5 \times v^2 = 12.5$

A1

Speed is $\sqrt{5} \text{ m s}^{-1}$
AG

A1

3

(c) Resolving vertically, $R = 5g$

B1

$$F = \mu R$$

M1

$$0.4 \times 5g = 2g$$

M1

Using change in energy = work done:
 $2g \times 0.5 =$

M1 for force $\times$ distance

M1

$$\frac{1}{2} \times 5 \times (\sqrt{5}^2) - \frac{1}{2} \times 5 \times v^2$$

A1 first term (or 12.5)

A1 second term (inc -)

A1,A1

$$9.8 = 12.5 - \frac{5}{2}v^2$$

$$v^2 = 1.08$$

Speed is 1.04 m s^{-1}

A1

7

[12]

Q10.

(a)
$$\text{EPE} = \frac{1}{2} \times \frac{30}{0.5} \times 1.3^2 = 50.7 \text{ J}$$

use of EPE formula

M1

correct EPE

A1

2

(b) (i) $50.7 = \frac{1}{2} \times 2v + \frac{1}{2} \times \frac{30}{0.5} \times 0.8^2$
three term energy equation
two terms correct

M1

all terms correct

A1

$50.7 = v^2 + 19.2$ AG

A1

solving for v

dM1

$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$
correct v from correct working

A1

5

(ii) $50.7 = \frac{1}{2} \times 2v^2$

M1

two term energy equation
correct equation

A1

$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$

correct velocity

A1

3

(c) $\frac{1}{2} \times 2v^2 = 50.7 - 1.8 \times 0.1 \times 2 \times 9.8$

finding friction force

M1

correct friction force

A1

three term energy equation

M1

correct equation

A1

$$v = \sqrt{47.172} = 6.87 \text{ ms}^{-1}$$

correct velocity

A1

5

[15]

Q11.

(a) $KE = \frac{1}{2} \times 0.6 \times 14^2 = 58.8 \text{ J}$

use of KE formula

M1

correct energy

A1

2

(b) $0.6 \times 9.8h = 58.8$

two term energy equation involving PE and previous energy

M1

$$h = \frac{58.8}{0.6 \times 9.8} = 10 \text{ m}$$

correct equation

A1

correct height

A1

3

Note: Constant acceleration methods not accepted.

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(c) (i) WD against resistance

three term energy equation

M1

$$= 58.8 - 0.6 \times 9.8 \times 8$$

correct equation

A1

$$= 11.76 = 11.8 \text{ J (to 3 sf)}$$

correct value

A1

3

(ii) $8F = 11.76$

using work done = Fd with d = 8

M1

$$F = 1.47 \text{ N}$$

correct force

accept 1.48

A1ft

2

- (d) The magnitude of the force would vary with the speed of the ball.

appropriate explanation

B1

1

[11]

Q12.

(a) (i) $\frac{1}{2} \times 28 \times 1^2 + 28 \times 9.8 \times 2.5 = \frac{1}{2} \times 28 \times v^2$
M1 all 3 terms – 1 each term correct

M1A2

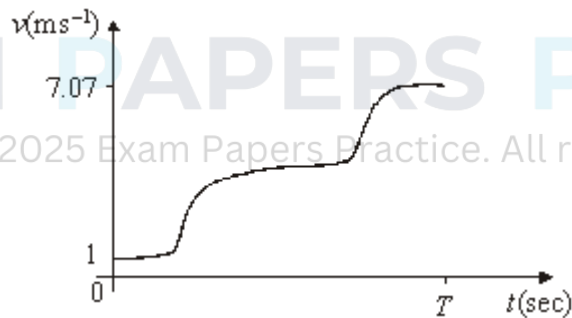
$$v = 7.07 \text{ ms}^{-1} \text{ (3 sf)}$$

Convincingly obtained

A1

4

(ii)



v increasing

B1

*accept straight line, not horizontal
 labels all correct (1, 7.07, T)*

B1

correct shape

B1

3

- (b) Initial energy = PE + KE

$$\frac{1}{2} \times 28 \times 1 + 28 \times 9.8 \times 2.5$$

M1

$$700 - \frac{1}{2} \times 28 \times v^2 = 350$$

M1 work/energy principle

A1 correct

M1A1

$$v = 5 \text{ ms}^{-1}$$

ft slip eg sign

A1F

4

[11]

Q13.

(a) (i) $\text{KE} = \frac{1}{2} \times 35 \times 2^2 = 70 \text{ J}$

KE calculation using $v = 2$

Correct KE

M1

A1

2

(ii) $\text{Gain in KE} = \frac{1}{2} \times 35 \times 6^2 - 70$

KE calculation using $v = 6$

M1

Correct expression for gain in KE

A1

$$= 560 \text{ J}$$

Correct gain

A1

3

(b) $\text{PE lost} = 35 \times 9.8 \times 10 \sin 40^\circ$

PE calculation with attempt to find height

M1

$$= 2200 \text{ J}$$

ag *Correct PE; allow 2204*

A1

2

(c) $2204 - 560 = 10F$

Energy lost = Fx including 560

M1

Correct equation

A1

$F = 164 \text{ N}$

Correct F

Alternative

M1: Three term equation of motion

A1: Correct equation

A1: Correct force

A1

3

(d) $\frac{1}{2} \times 35 \times 6^2 = Fs$

Use of KE lost = Fx

M1

Correct equation

A1ft

$s = \frac{630}{F} = 3.83 \text{ m}$

Correct length

Follow through

F Allow 3.84

Alternative

M1: Finding acceleration

A1: Correct use of a constant acceleration equation

A1: Correct length from correct working

A1ft

3

[13]

Q14.

(a) (i) $KE = 0.5 \times 3 \times 9.8 = 14.7 \text{ J}$

KE based on a fall of 3 m

cao

M1 A1

2

(ii) $KE = 0.5 \times 6 \times 9.8 = 29.4J$

Using 29.4

B1

$$\frac{1}{2} \times 0.5v^2 = 29.4$$

Forming and solving an energy equation

M1

$$v = \sqrt{\frac{29.4}{0.25}} = 10.8 \text{ ms}^{-1}$$

awrt 10.8

A1

3

(b) $\frac{1}{2} \times 0.5 \times 2^2 = 29.4 - 6R$

Three term energy equation with 6R

Correct equation

M1 A1

$$R = \frac{29.4 - 1}{6} = 4.73 \text{ N}$$

Solving for R

awrt 4.73

m1 A1

4

Examiner reports

Q2.

Part (a) of this question was answered successfully by most candidates. Although many candidates were successful in part (b), a few attempted to use kinetic energy. Relatively few candidates were successful in part (c), with most candidates stating that when the particle was at *B* the string was stretched and hence the string had elastic potential energy and thus the particle must move. The fact that the particle would only move again if the tension in the string was greater than the friction, found in part (b), was only considered by the better candidates. Those candidates who had answered part (b) correctly usually answered parts (d) and (e) correctly.

Q3.

In part (a) most candidates found the initial EPE correctly, but a number forgot to add on the initial KE to find the energy of the block at *A*. In part (b)(i), most candidates found the frictional force to be $0.4 \mu g$ but often failed to multiply this by 5.5 to obtain the work done by the frictional force. Other candidates used $v^2 = u^2 + 2as$, ignoring the fact that constant acceleration formulae cannot be used with elastic strings. In part (b)(ii), candidates often used the speed after rebounding to be half the speed of the block before the impact, but when finding the kinetic energy forgot to square the half, when calculating v^2 . The required equation on part (ii) included three terms, KE when leaving *B*, work done by friction and the EPE in the string when the block came to rest at *B*. Unfortunately many candidates forgot that the string had a non-zero EPE when the block was at *B*.

Q4.

Most candidates answered part (a) correctly. In part (b), a significant number did not read the question carefully to see that an *energy* method was required. Thus the *only* acceptable method was to use $Fs = \text{change in kinetic energy}$.

Q5.

Parts (a) and (b) of this question were answered well. In part (c), many complicated methods were used to find the work done; often speed was considered and it was common for the work done, exactly 48 J from part (a), to become a value fairly close to 48 J. Most candidates found the frictional force to be $5g\mu$, but relatively few candidates used work done by friction to be $5g\mu$ multiplied by the distance moved, which was 2 m.

Q6.

Candidates found this question straightforward and answered it well

Q7.

This proved to be a successful start for many candidates.

Q8.

Parts (a) and (b) were answered very well. Some omitted part (c) while others stopped after finding the change in energy.

Q9.

Parts (a) and (b) caused few problems for the vast majority of candidates. In part (c) it was common to see the frictional force to be $5g \times 0.4 = 19.6 \text{ N}$, but this was often not multiplied by the distance moved, 0.5 (m), to find the work done.

Q10.

Part (a) was done very well, with very few incorrect responses. Part (b) of the question was also done fairly well, with the printed answer helping some candidates. The main issue here was that some candidates did not use the correct distances to find the elastic potential energies. Part (c) was much more demanding. Many candidates were able to calculate the magnitude of the friction force correctly, but far fewer were able to incorporate this correctly into an energy equation.

Q11.

Parts (a) and (b) of this question were done very well by the vast majority of candidates. The number of candidates using constant acceleration equations instead of energy was quite small.

Part (c) was found more difficult. There were some candidates who did not really know how to proceed with the question, and others who were confused about how to calculate the work done against the air resistance. One of the most common incorrect responses was to produce the answer 47.04 J.

Part (d) was answered well by some candidates, but some gave fairly poor and very brief answers.

Q12.

This question also proved popular, with parts (a) and (b)(ii) showing a sound understanding of the principles of energy and work. There were many interesting solutions to part (b)(i), with the majority realising that the velocity of the particle increased at stages throughout the motion. Often candidates produced the correct shape in their sketch, but labelling of the axes was incomplete.

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Q13.

Parts (a) and (b) of this question were generally done very well by the vast majority of candidates. Parts (c) and (d) caused more problems for the candidates. Quite a number of candidates tried to use equations of motion and constant acceleration equations to answer parts (c) and (d). While some were able to do this successfully, many were not. In part (c), quite a few candidates simply assumed that $F = 35g \sin 40^\circ = 220$. *Candidates were sometimes more successful with part (d), but there were some who used 560 J for the kinetic energy at B instead of 630 J.*

Q14.

Part (a) of this question was done well by many candidates. A few used 14.7 instead of 29.4 when trying to find the speed. The main difficulty that the candidates experienced was with part (b). Here, many candidates found the work done against the resistance force, but did not attempt to go further to find the force. Several candidates gave the answer 28.4 N, which showed some confusion between force and energy.